Pharmacodynamics
Mechanism of drug action

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Objectives

- Know how a drug works
- List the different mechanisms by which a drug acts

- Pharmacodynamics is the study of drug effects.
- It starts with describing what the drugs do, and goes on to explain how they do it. Thus, it attempts to elucidate the complete action-effect sequence and the dose-effect relationship.
- Modification of the action of one drug by another drug is also an aspect of pharmacodynamics.

Pharmacodynamics

- Qualitative aspects: receptors, enzymes, selectivity.
- Quantitative aspects: dose response, potency, therapeutic efficacy, tolerance.

QUALITATIVE ASPECTS

- The starting point is to consider what drugs do and how they do it, i.e. the nature of drug action.
- The body functions through control systems that involve chemotransmitters or local hormones, receptors, enzymes, carrier molecules and other specialised macromolecules such as DNA.

- Most medicinal drugs act by altering the body's control systems and, in general, they do so by binding to some specialised constituent of the cell, selectively to alter its function and consequently that of the physiological or pathological system to which it contributes.
- Such drugs are structurally specific in that small modifications to their chemical structure may profoundly alter their effect.

PRINCIPLES OF DRUG ACTION

- Drugs (except those gene based) do not impart new functions to any system, organ or cell; they only alter the pace of ongoing activity.
- However, this alone can have profound medicinal as well as toxicological impact.
- The basic types of drug action can be broadly classed as:

1. Stimulation

- It refers to selective enhancement of the level of activity of specialized cells, e.g. adrenaline stimulates heart, pilocarpine stimulates salivary glands.
- However, excessive stimulation is often followed by depression of that function, e.g. high dose of picrotoxin, a central nervous system (CNS) stimulant, produces convulsions followed by coma and respiratory depression.

2. Depression

- It means selective diminution of activity of specialized cells, e.g. barbiturates depress CNS, quinidine depresses heart, omeprazole depresses gastric acid secretion.
- Certain drugs stimulate one type of cells but depress the other, e.g. acetylcholine stimulates intestinal smooth muscle but depresses SA node in heart.
- Thus, most drugs cannot be simply classed as stimulants or depressants.

3. Irritation

- This connotes a nonselective, often noxious effect and is particularly applied to less specialized cells (epithelium, connective tissue).
- Strong irritation results in inflammation, corrosion, necrosis and morphological damage.
- This may result in diminution or loss of function.

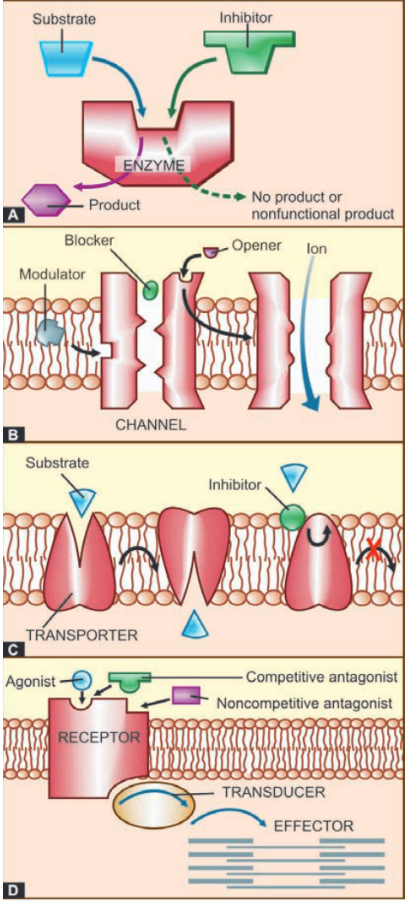
4. Replacement

- This refers to the use of natural metabolites, hormones or their congeners in deficiency states, e.g. levodopa in parkinsonism, insulin in diabetes mellitus, iron in anaemia.

5. Cytotoxic action

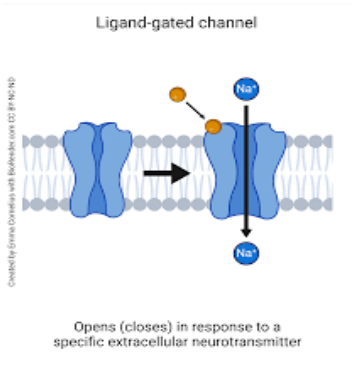
- Selective cytotoxic action on invading parasites or cancer cells, attenuating them without significantly affecting the host cells is utilized for cure/ palliation of infections and neoplasms, e.g. penicillin, chloroquine, zidovudine, cyclophosphamide, etc.

- Majority of drugs produce their effects by interacting with a discrete target biomolecule, which usually is a protein.
- Such mechanism confers selectivity of action to the drug.
- Functional proteins that are targets of drug action can be grouped into four major categories, which are enzymes, ion channels, transporters and receptors.
- However, a few drugs do act on other proteins (e.g. colchicine bind to the structural protein tubulin) or on nucleic acids (alkylating agents).

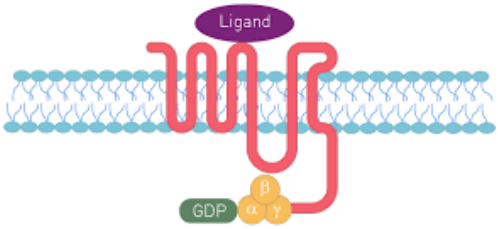


Mechanisms

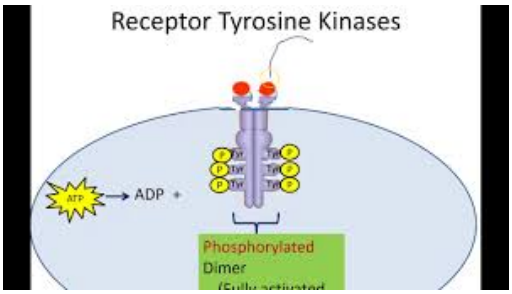
- An overview of the mechanisms of drug action shows that drugs act on specific receptors in the cell membrane and interior by:
- **Ligand-gated ion channels**, i.e. receptors coupled directly to membrane ion channels; neurotransmitters act on such receptors in the postsynaptic membrane of a nerve or muscle cell and give a response within <u>milliseconds</u>.



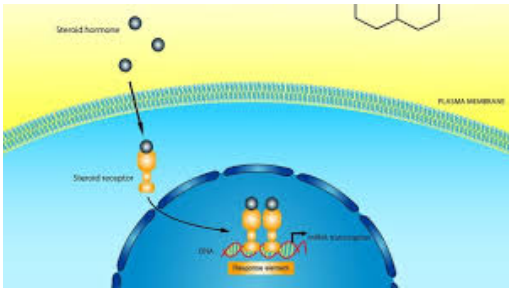
- G-protein-coupled receptor systems, i.e. receptors bound to the cell membrane and coupled to intracellular effector systems by a G-protein.
- For instance, catecholamines (the first messenger) activate b-adrenoceptors through a coupled G-protein system.
- This increases the activity of intracellular adenylyl cyclase, increasing the rate of formation of cyclic AMP (the second messenger), a modulator of the activity of several enzyme systems that cause the cell to act. The process takes seconds.



- **Protein kinase receptors**, so called because the structure incorporates a protein kinase, are targets for peptide hormones involved in the control of cell growth and differentiation, and the release of inflammatory mediators over a course of hours.



- **Cytosolic (nuclear) receptors**, i.e. within the cell itself, regulate DNA transcription and, thereby, protein synthesis, e.g. by steroid and thyroid hormones, a process that takes <u>hours or days</u>.



Drugs also act on processes within or near the cell by

- Enzyme inhibition, e.g. platelet cyclo-oxygenase by aspirin, cholinesterase by pyridostigmine, xanthine oxidase by allopurinol.
- Incorporation into larger molecules, e.g. 5-fluorouracil, an anticancer drug, is incorporated into messenger RNA in place of uracil.

- In the case of successful antimicrobial agents, altering metabolic processes unique to microorganisms, e.g. penicillin interferes with formation of the bacterial cell wall; or by showing enormous quantitative differences in affecting a process common to both humans and microbes, e.g. inhibition of folic acid synthesis by trimethoprim.

Outside the cell drugs act by:

- Direct chemical interaction, e.g. chelating agents, antacids.
- Osmosis, as with purgatives, e.g. magnesium sulphate, and diuretics, e.g. mannitol, which are active because neither they nor the water in which they are dissolved is absorbed by the cells lining the gut and kidney tubules respectively.

References

• KD-Tripathi-Pharmacology-Book.pdf

Further information:

• https://youtu.be/0Gl74JoNZi0?si=MFy470OuKHjB43Dw